

Digital Electronics Question and Answers

Combinational Circuits

Topic 5

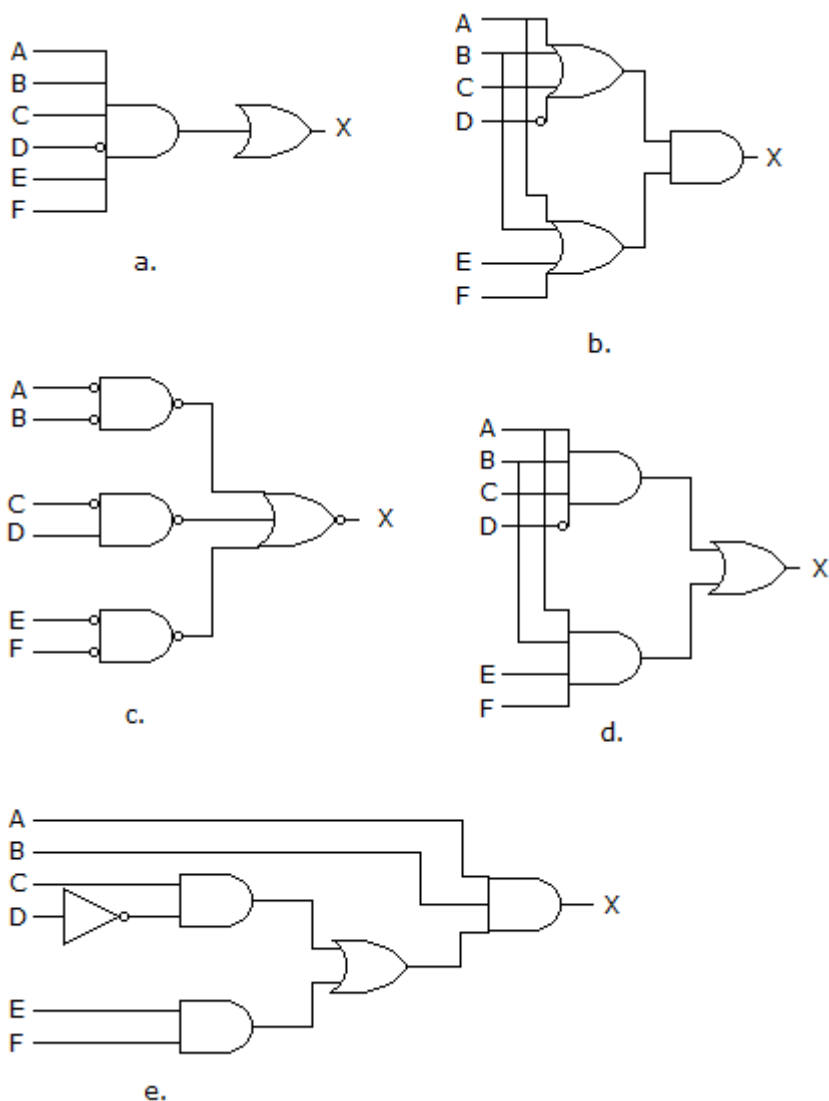
- 1.) combinational Circuits
- 2.) Multiplexers (Data Selectors)
- 3.) Demultiplexers (Data Distributors)
- 4.) Liquid Crystal Displays
- 5.) Encoders
- 6.) Parity Generators/Checkers
- 7.) Code Converters
- 8.) Magnitude Comparator

Digital Electronics Question and Answers

Combinational Circuits

1.) Combinational Circuits

1. Which of the circuits in figure (a to d) is the sum-of-products implementation of figure (e)?

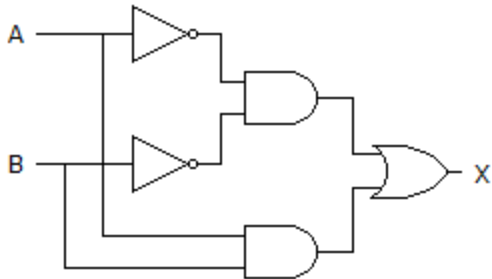


- a) a
- b) b
- c) c
- d) d

Answer: d

Explanation: SOP means Sum Of Products form which represents the sum of product terms having variables in complemented as well as in uncomplemented form. Here, the diagram of d contains the OR gate followed by the AND gates, so it is in SOP form.

2. Which of the following logic expressions represents the logic diagram shown?



- a) $X = AB' + A'B$
- b) $X = (AB)' + AB$
- c) $X = (AB)' + A'B'$
- d) $X = A'B' + AB$

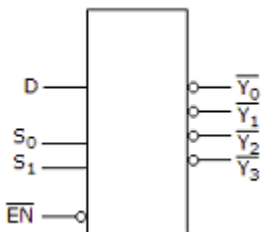
Answer: d

Explanation: 1st output of AND gate is $= A'B'$

2nd AND gate's output is $= AB$ and,

OR gate's output is $= (A'B') + (AB) = AB + A'B'$.

3. The device shown here is most likely a _____

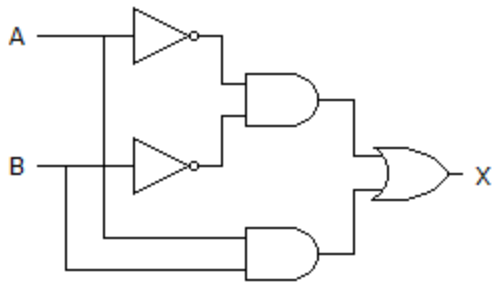


- a) Comparator
- b) Multiplexer
- c) Inverter
- d) Demultiplexer

Answer: d

Explanation: The given diagram is demultiplexer, because it takes single input & gives many outputs. A demultiplexer is a combinational circuit that takes a single output and latches it to multiple outputs depending on the select lines.

4. What type of logic circuit is represented by the figure shown below?

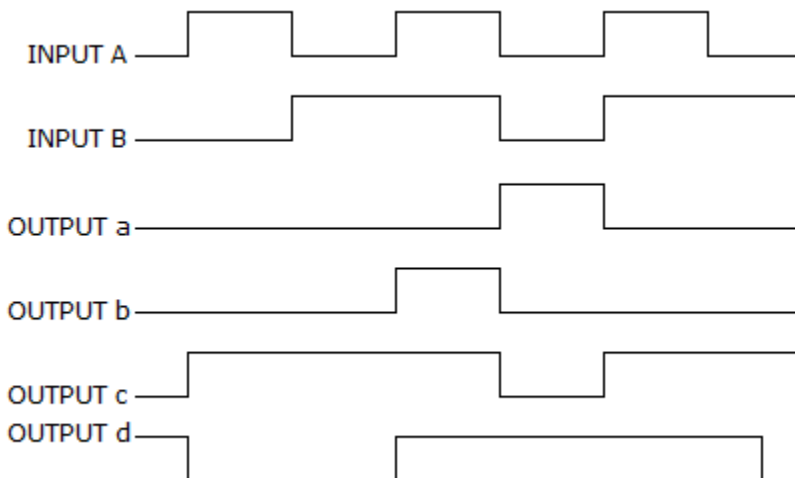


- a) XOR
- b) XNOR
- c) AND
- d) XAND

Answer: b

Explanation: After solving the circuit we get $(A'B') + AB$ as output, which is XNOR operation. Thus, it will produce 1 when inputs are even number of 1s or all 0s, and produce 0 when input is odd number of 1s.

5. For a two-input XNOR gate, with the input waveforms as shown below, which output waveform is correct?



- a) d
- b) a
- c) c
- d) b

Answer: a

Explanation: When both inputs are same then the o/p is high for a XNOR gate.

i.e., A B O/P

0 0 1

0 1 0

1 0 0

1 1 1.

Thus, it will produce 1 when inputs are even number of 1s or all 0s, and produce 0 when input is odd number of 1s.

6. Which of the following combinations of logic gates can decode binary 1101?

- a) One 4-input AND gate
- b) One 4-input AND gate, one inverter
- c) One 4-input AND gate, one OR gate
- d) One 4-input NAND gate, one inverter

Answer: b

Explanation: For decoding any number output must be high for that code and this is possible in One 4-input AND gate, one inverter option only. We can have 1st, 2nd and 4th bit high and supplied to the AND gate and the 3rd bit (low) going through inverter 1st and then going to the AND gate. A decoder is a combinational circuit that converts binary data to n-coded data upto 2^n outputs.

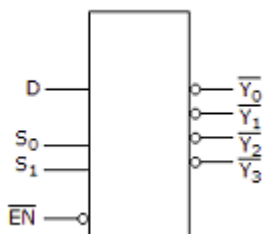
7. What is the indication of a short to ground in the output of a driving gate?

- a) Only the output of the defective gate is affected
- b) There is a signal loss to all load gates
- c) The node may be stuck in either the HIGH or the LOW state
- d) The affected node will be stuck in the HIGH state

Answer: b

Explanation: Short to ground in the output of a driving gate indicates of a signal loss to all load gates. This results in information being disrupted and loss of data.

8. For the device shown here, assume the D input is LOW, both S inputs are LOW and the input is LOW. What is the status of the Y' outputs?



- a) All are HIGH
- b) All are LOW
- c) All but Y0 are LOW
- d) All but Y0 are HIGH

Answer: d

Explanation: In the given diagram, S0 and S1 are selection bits. So,
I/P S0 S1 O/P

D = 0 0 0 Y0

D = 0 0 1 Y1

D = 0 1 0 Y2

D = 0 1 1 Y3

Hence, inputs are S0 and S1 are Low means 0, so output is Y0 and rest all are HIGH.

9. The carry propagation can be expressed as _____

- a) $C_p = AB$
- b) $C_p = A + B$
- c) All but Y0 are LOW
- d) All but Y0 are HIGH

Answer: b

Explanation: This happens in parallel adders (where we try to add numbers in parallel via more than one adders). A carry propagation occurs when carry from one adder needs to be forwarded to other adder and that second adder is holding the computation (addition) because carry from first adder has not come yet. So, there is a slight delay for second adder and this is known as carry propagation.

10. 3 bits full adder contains _____

- a) 3 combinational inputs
- b) 4 combinational inputs
- c) 6 combinational inputs
- d) 8 combinational inputs

Answer: d

Explanation: Full Adder is a combinational circuit with 3 input bits and 2 output bits CARRY and SUM. Three bits full adder requires $2^3 = 8$ combinational circuits.

2.) Multiplexers (Data Selectors)

1. What is a multiplexer?

- a) It is a type of decoder which decodes several inputs and gives one output
- b) A multiplexer is a device which converts many signals into one
- c) It takes one input and results into many output
- d) It is a type of encoder which decodes several inputs and gives one output

Answer: b

Explanation: A multiplexer (or MUX) is a device that selects one of several analog or digital input signals and forwards the selected input into a single line, depending on the active select lines.

2. Which combinational circuit is renowned for selecting a single input from multiple inputs & directing the binary information to output line?

- a) Data Selector
- b) Data distributor
- c) Both data selector and data distributor
- d) DeMultiplexer

Answer: a

Explanation: Data Selector is another name of Multiplexer. A multiplexer (or MUX) is a device that selects one of several analog or digital input signals and forwards the selected input into a single line, depending on the active select lines.

3. It is possible for an enable or strobe input to undergo an expansion of two or more MUX ICs to the digital multiplexer with the proficiency of large number of _____

- a) Inputs
- b) Outputs
- c) Selection lines
- d) Enable lines

Answer: a

Explanation: It is possible for an enable or strobe input to undergo an expansion of two or more MUX ICs to the digital multiplexer with the proficiency of large number of inputs.

4. Which is the major functioning responsibility of the multiplexing combinational circuit?

- a) Decoding the binary information
- b) Generation of all minterms in an output function with OR-gate
- c) Generation of selected path between multiple sources and a single destination
- d) Encoding of binary information

Answer: c

Explanation: The major functioning responsibility of the multiplexing combinational circuit is generation of selected path between multiple sources and a single destination because it makes the circuit too flexible. A multiplexer (or MUX) is a device that selects one of several analog or digital input signals and forwards the selected input into a single line, depending on the active select lines.

5. What is the function of an enable input on a multiplexer chip?

- a) To apply Vcc
- b) To connect ground
- c) To active the entire chip
- d) To active one half of the chip

Answer: c

Explanation: Enable input is used to active the chip, when enable is high the chip works (ACTIVE), when enable is low the chip does not work (MEMORY). However, Enable can be Active-High or Active-Low, indicating it is active either when it is connected to VCC or GND respectively.

6. One multiplexer can take the place of _____

- a) Several SSI logic gates
- b) Combinational logic circuits
- c) Several Ex-NOR gates
- d) Several SSI logic gates or combinational logic circuits**

Answer: d

Explanation: A multiplexer (or MUX) is a device that selects one of several analog or digital input signals and forwards the selected input into a single line, depending on the active select lines. Since many operational behaviour can be performed by using a multiplexer. Whereas, a combinational circuit is a combination of many logic gates which makes the circuit more complex.

7. A digital multiplexer is a combinational circuit that selects _____

- a) One digital information from several sources and transmits the selected one
- b) Many digital information and convert them into one
- c) Many decimal inputs and transmits the selected information
- d) Many decimal outputs and accepts the selected information

Answer: a

Explanation: A digital multiplexer is a combinational circuit that selects one digital information from several sources and transmits the selected information on a single output line depending on the status of the select lines. That is why it is also known as a data selector.

8. In a multiplexer, the selection of a particular input line is controlled by _____

- a) Data controller
- b) Selected lines
- c) Logic gates
- d) Both data controller and selected lines

Answer: b

Explanation: The selection of a particular input line is controlled by a set of selected lines in a multiplexer, which helps to select a particular input from several sources.

9. If the number of n selected input lines is equal to 2^m then it requires _____ select lines.

- a) 2
- b) m
- c) n
- d) 2^n

Answer: b

Explanation: If the number of n selected input lines is equal to 2^m then it requires m select lines to select one of m select lines.

10. How many select lines would be required for an 8-line-to-1-line multiplexer?

- a) 2
- b) 4
- c) 8
- d) 3

Answer: d

Explanation: 2^n input lines, n control lines and 1 output line available for MUX. Here, 8 input lines mean 2^3 inputs. So, 3 control lines are possible. Depending on the status of the select lines, the input is selected and fed to the output.

11. A basic multiplexer principle can be demonstrated through the use of a _____

- a) Single-pole relay
- b) DPDT switch
- c) Rotary switch
- d) Linear stepper

Answer: c

Explanation: A basic multiplexer principle can be demonstrated through the use of a rotary switch. Since its behaviour is similar to the multiplexer. There are around 10 digits out of which one is selected one at a time and fed to the output.

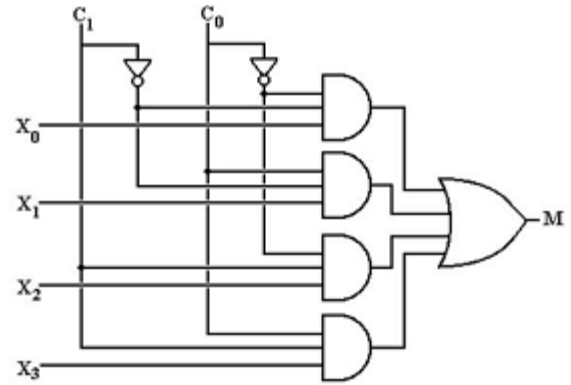
12. How many NOT gates are required for the construction of a 4-to-1 multiplexer?

- a) 3
- b) 4
- c) 2
- d) 5

Answer: c

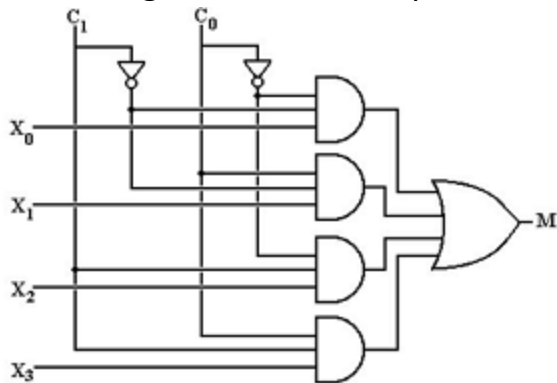
Explanation: There are two NOT gates required for the construction of 4-to-1 multiplexer.

x_0 , x_1 , x_2 and x_3 are the inputs and C_1 and C_0 are the select lines and M is the output.



The diagram of a 4-to-1 multiplexer is shown below:

13. In the given 4-to-1 multiplexer, if $c_1 = 0$ and $c_0 = 1$ then the output M is _____



- a) x_0
- b) x_1
- c) x_2
- d) x_3

Answer: b

Explanation: The output will be x_1 , because $c_1 = 0$ and $c_0 = 1$ results into 1 which further results as x_1 . And rest of the AND gates gives output as 0.

14. The enable input is also known as _____

- a) Select input
- b) Decoded input
- c) Strobe
- d) Sink

Answer: c

Explanation: The enable input is also known as strobe which is used to cascade two or more multiplexer ICs to construct a multiplexer with a larger number of inputs. Enable input activates the multiplexer to operate.

15. 4 to 1 MUX would have _____

- a) 2 inputs

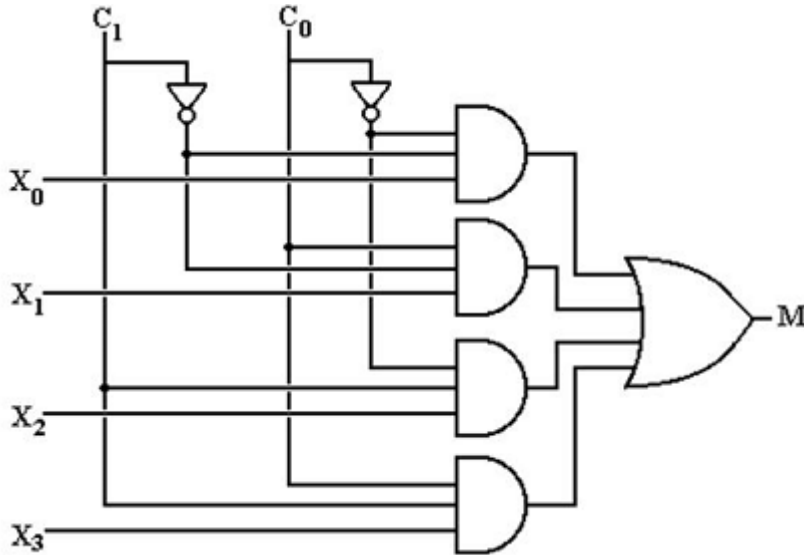
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- b) 3 inputs
- c) 4 inputs
- d) 5 inputs

Answer: c

Explanation: 4 to 1 multiplexer would have 4 inputs (X_0, X_1, X_2, X_3), 2 select lines (C_1, C_0) and 1 output (M). It can be observed from this diagram:



16. The two input MUX would have _____

- a) 1 select line
- b) 2 select lines
- c) 4 select lines
- d) 3 select lines

Answer: a

Explanation: The two input multiplexer would have n select lines in 2^n . Thus $n = 1$. Therefore, it has 1 select line.

17. A combinational circuit that selects one from many inputs are _____

- a) Encoder
- b) Decoder
- c) Demultiplexer
- d) Multiplexer

Answer: d

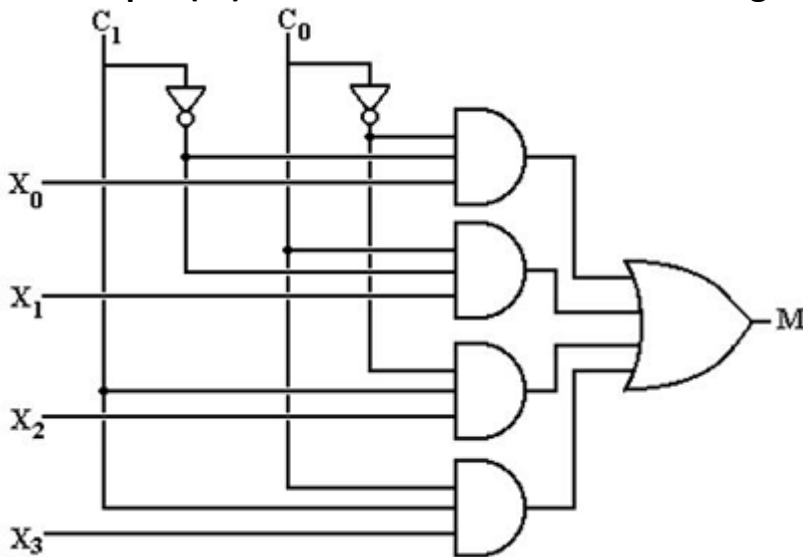
Explanation: A combinational circuit that selects one from many inputs is known as Multiplexer. Whereas, a combinational circuit that divides one input into multiple outputs is known as Demultiplexer.

18. 4 to 1 MUX would have _____

- a) 1 output
- b) 2 outputs
- c) 3 outputs
- d) 4 outputs

Answer: a

Explanation: 4 to 1 multiplexer would have 4 inputs (X_0, X_1, X_2, X_3), 2 select lines (C_1, C_0) and 1 output (M). It can be observed from this diagram:



19. Which of the following circuit can be used as parallel to serial converter?

- a) Multiplexer
- b) Demultiplexer
- c) Decoder
- d) Digital counter

Answer: a

Explanation: A combinational circuit that selects one from many inputs is known as **Multiplexer**. In multiplexer, different inputs are inserted parallelly and then it gives one output which is in serial form.

20. A combinational circuit is one in which the output depends on the _____

- a) Input combination at the time
- b) Input combination and the previous output
- c) Input combination at that time and the previous input combination
- d) Present output and the previous output

Answer: a

Explanation: A combinational circuit is one in which the output depends on the input

combination at the time, whereas, a sequential circuit is one in which the output depends on present input as well past outputs.

21. Without any additional circuitry an 8:1 MUX can be used to obtain _____

- a) Some but not all Boolean functions of 3 variables
- b) All function of 3 variables but none of 4 variables
- c) All functions of 3 variables and some but not all of 4 variables
- d) All functions of 4 variables

Answer: d

Explanation: A $2^n:1$ MUX can implement all logic functions of $(n+1)$ variables without any additional circuitry. Thus 8:1 MUX can implement all logic functions of $(3+1)$ variables, for 4 variables there are 16 possible combinations. So to use 8:1 MUX use 3 inputs as select lines of MUX and the 4th input as input of MUX.

22. A basic multiplexer principle can be demonstrated through the use of a _____

- a) Single-pole relay
- b) DPDT switch
- c) Rotary switch
- d) Linear stepper

Answer: c

Explanation: A combinational circuit that selects one from many inputs is known as Multiplexer. A basic multiplexer principle can be demonstrated through the use of a rotary switch. Because rotary switch gives one output corresponding to their inputs.

22. One multiplexer can take the place of _____

- a) Several SSI logic gates
- b) Combinational logic circuits
- c) Several Ex-NOR gates
- d) Several SSI logic gates or combinational logic circuits

Answer: d

Explanation: A combinational circuit that selects one from many inputs is known as Multiplexer. One multiplexer can take the place of several SSI logic gates or combinational logic circuits because it has a lot of functions to perform different operations.

22. The inputs/outputs of an analog multiplexer/demultiplexer are _____

- a) Bidirectional
- b) Unidirectional
- c) Even parity
- d) Binary-coded decimal

Answer: a

Explanation: One multiplexer can be used as demultiplexer. Hence, it is called bidirectional or two-way transmission.

23. If enable input is high then the multiplexer is _____

- a) Enable
- b) Disable
- c) Saturation
- d) High Impedance

Answer: b

Explanation: If enable input is high then the multiplexer is disabled because enable input is in inverted mode always (i.e. E').

24. What is data routing in a multiplexer?

- a) It spreads the information to the control unit
- b) It can be used to route data from one of several source to destination
- c) It is an application of multiplexer
- d) It can be used to route data and it is an application of multiplexer

Answer: d

Explanation: Multiplexing means passing more than one data through the same channel. Data routing is an application of multiplexer and it can be used to route data from one of several source to destination.

3.) Demultiplexers (Data Distributors)

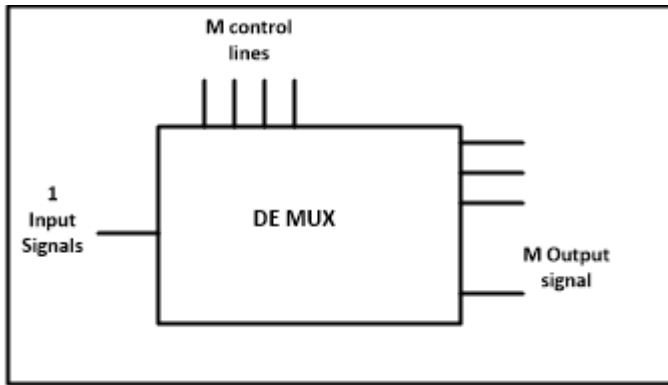
1. The word demultiplex means _____

- a) One into many
- b) Many into one
- c) Distributor
- d) One into many as well as Distributor

Answer: d

Explanation: The word demultiplex means “one into many” and distributor. A demultiplexer sends a single input to multiple outputs, depending on the select lines. It is clear from the

diagram:



2. Why is a demultiplexer called a data distributor?

- a) The input will be distributed to one of the outputs
- b) One of the inputs will be selected for the output
- c) The output will be distributed to one of the inputs
- d) Single input to Single Output

Answer: a

Explanation: A demultiplexer sends a single input to multiple outputs, depending on the select lines. For one input, the demultiplexer gives several outputs. That is why it is called a data distributor.

3. Most demultiplexers facilitate which type of conversion?

- a) Decimal-to-hexadecimal
- b) Single input, multiple outputs
- c) AC to DC
- d) Odd parity to even parity

View Answer

Answer: b

Explanation: A demultiplexer sends a single input to multiple outputs, depending on the select lines. Demultiplexer converts single input into multiple outputs.

4. In 1-to-4 demultiplexer, how many select lines are required?

- a) 2
- b) 3
- c) 4
- d) 5

Answer: a

Explanation: The formula for total no. of outputs is given by 2^n , where n is the no. of select lines. Therefore, for 1:4 demultiplexer, 2 select lines are required.

5. In a multiplexer the output depends on its _____

- a) Data inputs
- b) Select inputs
- c) Select outputs
- d) Enable pin

Answer: b

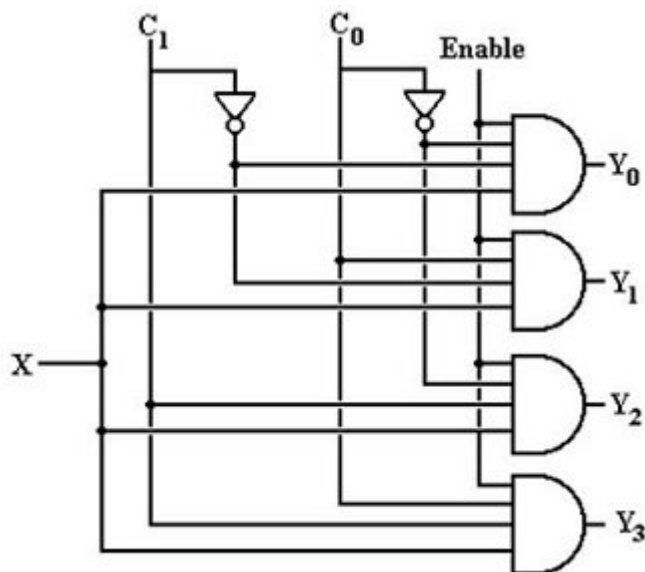
Explanation: A demultiplexer sends a single input to multiple outputs, depending on the select lines. As the select input changes, the output of the multiplexer varies according to that input.

6. In 1-to-4 multiplexer, if $C_1 = 0$ & $C_2 = 1$, then the output will be _____

- a) Y_0
- b) Y_1
- c) Y_2
- d) Y_3

Answer: b

Explanation: It can be calculated from the figure shown below:



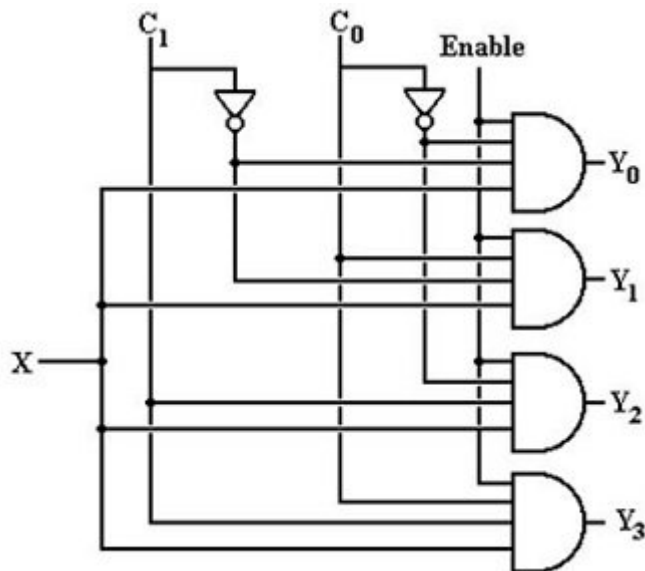
For $C_0 = 1$ and $C_1 = 0$, Y_1 will be the output as 0 and 1 are the bit combinations of 1.

7. In 1-to-4 multiplexer, if $C_1 = 1$ & $C_2 = 1$, then the output will be _____

- a) Y_0
- b) Y_1
- c) Y_2
- d) Y_3

Answer: d

Explanation: It can be calculated from the figure shown below:



For $C_0 = 1$ and $C_1 = 0$, Y_3 will be the output as 0 and 1 are the bit combinations of 1.

8. How many select lines are required for a 1-to-8 demultiplexer?

- a) 2
- b) 3
- c) 4
- d) 5

Answer: b

Explanation: The formula for total no. of outputs is given by 2^n , where n is the no. of select lines. In this case $n = 3$ since $2^3 = 8$.

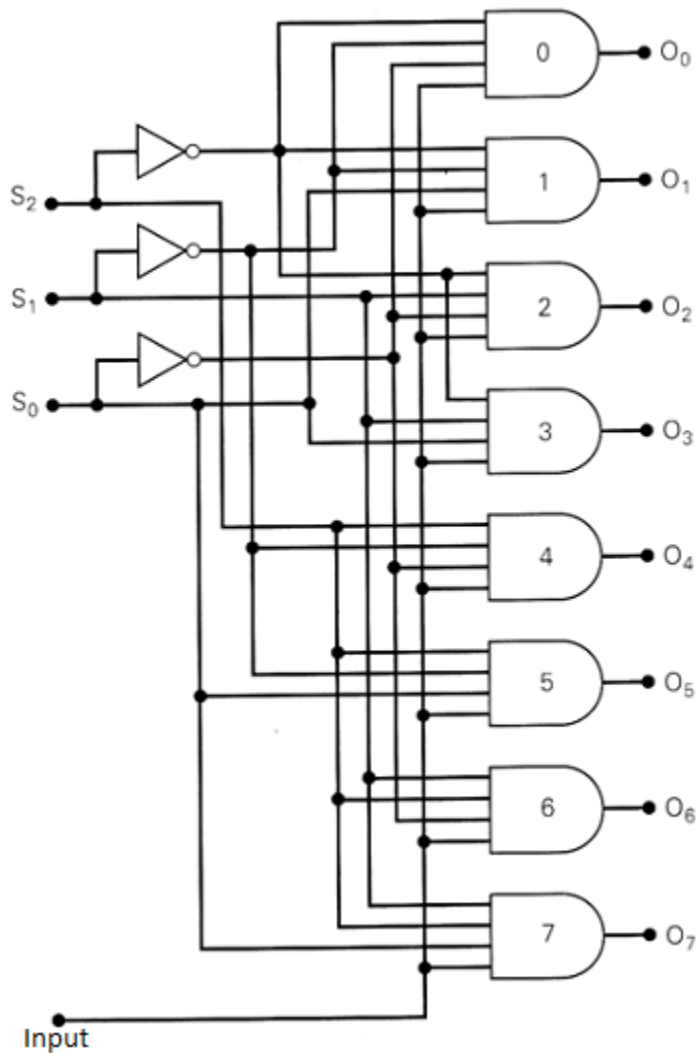
9. How many AND gates are required for a 1-to-8 multiplexer?

- a) 2
- b) 6
- c) 8
- d) 5

Answer: c

Explanation: The number of AND gates required will be equal to the number of outputs in a demultiplexer, which are 8.

10. The output Q4 of this 1-to-8 demultiplexer is _____



- a) $Q2.(Q1)'.Q0.I$
- b) $Q2.Q1.(Q0)'.I$
- c) $Q2.(Q1)'.(Q0)'.I$
- d) $Q2.(Q1).Q0.I$

Answer: c

Explanation: The logic gates for Q4 implement the function $Q2.(Q1)'.(Q0)'$. This means: Q2 should be high (1).

Q1 should be low (0) – denoted by the ' symbol after Q1.

Q0 should be low (0) – denoted by the ' symbol after Q0.

When the input I is high (1), the output Q4 will also be high (1) because all the conditions for Q4 are met (Q2 is high, Q1 is low, and Q0 is low).

Therefore, the output Q4 of the demultiplexer is correctly represented by $Q2.(Q1)'.(Q0)'.I$.

11. Which IC is used for the implementation of 1-to-16 DEMUX?

- a) IC 74154
- b) IC 74155

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c) IC 74139

d) IC 74138

Answer: a

Explanation: IC 74154 is used for the implementation of 1-to-16 DEMUX, whose output is inverted input.

12. Why is a demultiplexer called a data distributor?

a) The input will be distributed to one of the outputs

b) One of the inputs will be selected for the output

c) The output will be distributed to one of the inputs

d) Single input gives single output

Answer: a

Explanation: A demultiplexer sends a single input to multiple outputs, depending on the select lines. For one input, the demultiplexer gives several outputs. That is why it is called a data distributor.

13. Most demultiplexers facilitate which type of conversion?

a) Decimal-to-hexadecimal

b) Single input, multiple outputs

c) AC to DC

d) Odd parity to even parity

Answer: b

Explanation: A demultiplexer sends a single input to multiple outputs, depending on the select lines. Demultiplexer converts single input into multiple outputs.

14. In 1-to-4 demultiplexer, how many select lines are required?

a) 2

b) 3

c) 4

d) 5

Answer: a

Explanation: The formula for total no. of outputs is given by 2^n , where n is the no. of select lines. Therefore, for 1:4 demultiplexer, 2 select lines are required.

15. In a multiplexer the output depends on its _____

a) Data inputs

b) Select inputs

c) Select outputs

d) Enable pin

Answer: b

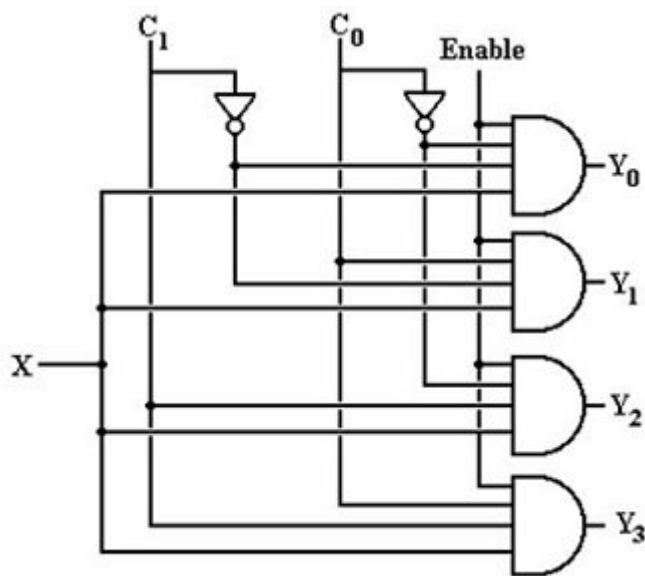
Explanation: A demultiplexer sends a single input to multiple outputs, depending on the select lines. As the select input changes, the output of the multiplexer varies according to that input.

16. In 1-to-4 multiplexer, if $C_1 = 1$ & $C_2 = 1$, then the output will be _____

- a) Y_0
- b) Y_1
- c) Y_2
- d) Y_3

Answer: d

Explanation: It can be calculated from the figure shown below:



For $C_0 = 1$ and $C_1 = 1$, Y_3 will be the output as 0 and 1 are the bit combinations of 1.

17. How many select lines are required for a 1-to-8 demultiplexer?

- a) 2
- b) 3
- c) 4
- d) 5

Answer: b

Explanation: The formula for total no. of outputs is given by 2^n , where n is the no. of select lines. In this case $n = 3$ since $2^3 = 8$.

18. How many AND gates are required for a 1-to-8 multiplexer?

- a) 2
- b) 6
- c) 8
- d) 5

Answer: c

Explanation: The number of AND gates required will be equal to the number of outputs in a demultiplexer, which are 8.

19. Which IC is used for the implementation of 1-to-16 DEMUX?

- a) IC 74154
- b) IC 74155
- c) IC 74139
- d) IC 74138

Answer: a

Explanation: IC 74154 is used for the implementation of 1-to-16 DEMUX, whose output is inverted input.

4.) Liquid Crystal Displays

1. The full form of LCD is _____

- a) Liquid Crystal Display
- b) Liquid Crystalline Display
- c) Logical Crystal Display
- d) Logical Crystalline Display

Answer: a

Explanation: The full form of LCD is “Liquid Crystal Display”. They provide thinner displays as compared to Cathode Ray Tubes.

2. The optical properties of liquid crystals depend on the direction of _____

- a) Air
- b) Solid
- c) Light
- d) Water

Answer: c

Explanation: The optical properties of liquid crystals depend on the direction of light travels through a layer of the material.

3. By which properties, the orientation of molecules in a layer of liquid crystals can be changed?

- a) Magnetic field
- b) Electric field
- c) Electromagnetic field
- d) Gallois field

Answer: b

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Explanation: In LCD, the electric field is induced by a small electric voltage applied across it; Due to which the orientation of molecules in a layer of liquid crystals can be changed.

4. Electro-optical effect is produced in _____

- a) LED
- b) LCD
- c) OFC
- d) OLED

Answer: b

Explanation: An electric field (induced by a small electric voltage) can change the orientation of molecules in a layer of liquid crystal and thus affect its optical properties. Such a process is termed an electro-optical effect, and it forms the basis for LCDs.

5. The direction of electric field in an LCD is determined by _____

- a) the molecule's chemical structure
- b) Crystalline surface structure
- c) Molecular Orbital Theory
- d) Quantum Cellular Automata

Answer: a

Explanation: For LCDs, the change in optical properties results from orienting the molecular axes either along or perpendicular to the applied electric field, the preferred direction being determined by the details of the molecule's chemical structure.

6. The first LCDs became commercially available in _____

- a) 1950s
- b) 1980s
- c) 1960s
- d) 1970s

Answer: c

Explanation: The first LCDs became commercially available in the late 1960s and were based on a light-scattering effect known as the dynamic scattering mode.

7. LCDs operate from a voltage ranges from _____

- a) 3 to 15V
- b) 10 to 15V
- c) 10V
- d) 5V

Answer: a

Explanation: LCDs operate from a voltage ranges from 3 to 15V rms. They provide thinner displays as compared to Cathode Ray Tubes.

8. LCDs operate from a frequency ranges from _____

- a) 10Hz to 60Hz
- b) 50Hz to 70Hz
- c) 30Hz to 60Hz
- d) None of the Mentioned

Answer: c

Explanation: LCDs operate from a frequency ranges from 30Hz to 60Hz. LCDs operate from a voltage ranges from 3 to 15V rms. They provide thinner displays as compared to Cathode Ray Tubes.

9. In 7 segment display, how many LEDs are used?

- a) 8
- b) 7
- c) 10
- d) 9

Answer: b

Explanation: There are 7 LEDs used in a 7 segment display. 7 segment displays are used for displaying decimal numerals which are comparatively convenient to dot matrix displays.

10. What is the backplane in LCD?

- a) The ac voltage applied between segment and a common element
- b) The dc voltage applied between segment and a common element
- c) The amount of power consumed
- d) For adjusting the intensity of the LCD

Answer: a

Explanation: The ac voltage applied between the segment and a common element is called the backplane(bp). In which each segment is driven by an EX-OR gate.

5.) Encoders

1. How many inputs will a decimal-to-BCD encoder have?

- a) 4
- b) 8
- c) 10
- d) 16

Answer: c

Explanation: An encoder is a combinational circuit encoding the information of 2^n input lines to n output lines, thus producing the binary equivalent of the input. Thus, a Decimal-to-bcd converter has decimal values as inputs which range from 0-9. So, a total of 10 inputs are there in a decimal-to-BCD encoder.

2. How many outputs will a decimal-to-BCD encoder have?

- a) 4
- b) 8
- c) 12
- d) 16

Answer: a

Explanation: An encoder is a combinational circuit encoding the information of 2^n input lines to n output lines, thus producing the binary equivalent of the input. Thus, a decimal to BCD encoder has 4 outputs.

3. How is an encoder different from a decoder?

- a) The output of an encoder is a binary code for 1-of-N input
- b) The output of a decoder is a binary code for 1-of-N input
- c) The output of an encoder is a binary code for N-of-1 output
- d) The output of a decoder is a binary code for N-of-1 output

Answer: a

Explanation: An encoder is a combinational circuit encoding the information of 2^n input lines to n output lines, thus producing the binary equivalent of the input. It performs the opposite operation of a decoder which results in 2^n outputs from n inputs. Thus, an encoder different from a decoder because of the output of an encoder is a binary code for 1-of-N input.

4. If we record any music in any recorder, such types of process is called _____

- a) Multiplexing
- b) Encoding
- c) Decoding
- d) Demultiplexing

Answer: b

Explanation: If we record any music in any recorder, it means that we are giving data to a recorder. So, such process is called encoding. Getting back the music from the recorded data is known as decoding.

5. Can an encoder be a transducer?

- a) Yes
- b) No
- c) May or may not be
- d) Both are not even related slightly

Answer: a

Explanation: Of course, a transducer is a device that has the capability to emit data as well as to accept. Transducer converts signal from one form of energy to another.

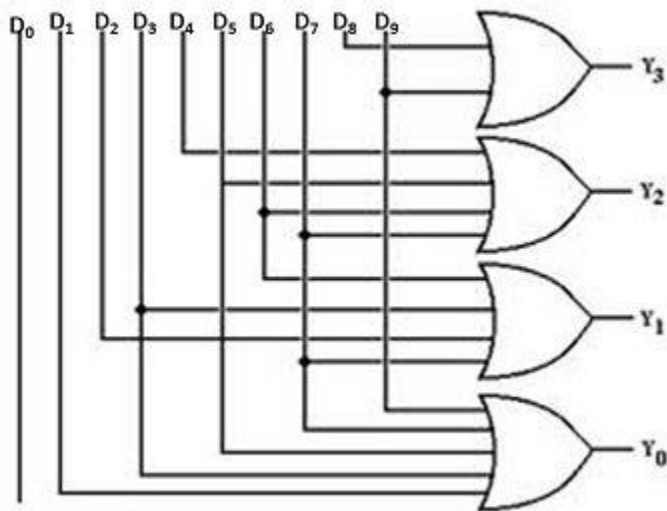
6. How many OR gates are required for a Decimal-to-bcd encoder?

- a) 2
- b) 10
- c) 3
- d) 4

Answer: d

Explanation: An encoder is a combinational circuit encoding the information of 2^n input lines to n output lines, thus producing the binary equivalent of the input.

This is clear from the diagram that it requires 4 OR gates:



7. How many OR gates are required for an octal-to-binary encoder?

- a) 3
- b) 2
- c) 8
- d) 10

Answer: a

Explanation: An encoder is a combinational circuit encoding the information of 2^n input lines to n output lines, thus producing the binary equivalent of the input. Thus, in octal to binary encoder there are 8 ($=2^3$) inputs, thus 3 output lines.

8. For 8-bit input encoder how many combinations are possible?

- a) 8
- b) 2^8
- c) 4
- d) 2^4

Answer: b

Explanation: An encoder is a combinational circuit encoding the information of 2^n input lines to n output lines, thus producing the binary equivalent of the input. There are

2^8 combinations are possible for an 8-bit input encoder but out of which only 8 are used using 3 output lines. It is a disadvantage of encoder.

9. The discrepancy of 0 output due to all inputs being 0 or D0, being 0 is resolved by using additional input known as _____

- a) Enable
- b) Disable
- c) Strobe
- d) Clock

Answer: a

Explanation: Such problems are resolved by using enable input, which behaves as active if it gets 0 as input since it is an active-low pin.

10. Can an encoder be called a multiplexer?

- a) No
- b) Yes
- c) Sometimes
- d) Never

Answer: b

Explanation: A multiplexer or MUX is a combination circuit that contains more than one input line, one output line and more than one selection line. Whereas, an encoder is also considered a type of multiplexer but without a single output line and without any selection lines.

11. If two inputs are active on a priority encoder, which will be coded on the output?

- a) The higher value
- b) The lower value
- c) Neither of the inputs
- d) Both of the inputs

Answer: a

Explanation: An encoder is a combinational circuit encoding the information of 2^n input lines to n output lines, thus producing the binary equivalent of the input. If two inputs are active on a priority encoder, the input of higher value will be coded in the output.

6.) Parity Generators/Checkers

1. How many outputs are present in a BCD decoder?

- a) 4
- b) 5

- c) 15
- d) 10

Answer: d

Explanation: A binary decoder is a combinational logic circuit which decodes binary information from n -inputs to a maximum of 2^n outputs. A BCD to Decimal decoder has 10 number of outputs because the decimal digit's range is from 0 to 9.

2. Which digital system translates coded characters into a more useful form?

- a) Encoder
- b) Display
- c) Counter
- d) Decoder

Answer: d

Explanation: A binary decoder is a combinational logic circuit which decodes binary information from n -inputs to a maximum of 2^n outputs. Decoder converts the coded characters into our required data form.

3. What control signals may be necessary to operate a 1-line-to-16 line decoder?

- a) Flasher circuit control signal
- b) A LOW on all gate enable inputs
- c) Input from a hexadecimal counter
- d) A HIGH on all gate enable circuits

Answer: b

Explanation: A LOW on all gate enable inputs is necessary to operate a 1-line-to-16 line decoder because enable pins are usually, active-low pins.

4. How many inputs are required for a 1-of-10 BCD decoder?

- a) 4
- b) 8
- c) 10
- d) 2

Answer: a

Explanation: A binary decoder is a combinational logic circuit which decodes binary information from n -inputs to a maximum of 2^n outputs. Therefore, for a BCD to decimal decoder, No. of inputs = 4 such that number of outputs is $\leq 2^n$.

5. A BCD decoder will have how many rows in its truth table?

- a) 10
- b) 9

- c) 8
- d) 3

Answer: a

Explanation: A binary decoder is a combinational logic circuit which decodes binary information from n -inputs to a maximum of 2^n outputs. Thus, BCD decoder will have 10 rows as it's input ranges from 0 to 9.

6. How many possible outputs would a decoder have with a 6-bit binary input?

- a) 32
- b) 64
- c) 128
- d) 16

Answer: b

Explanation: The possible outputs would be: $2^n = 64$ (Since $n = 6$ here).

7. Which is the way to convert BCD to binary using the hardware approach?

- a) By using MSI IC circuits
- b) By using a keyboard encoder
- c) By using an ALU
- d) By using UART

Answer: a

Explanation: One way to convert BCD to binary using the hardware approach is MSI (medium scale integration) IC circuits.

8. How many inputs are required for a 1-of-16 decoder?

- a) 2
- b) 16
- c) 8
- d) 4

Answer: d

Explanation: A binary decoder is a combinational logic circuit which decodes binary information from n -inputs to a maximum of 2^n outputs. Here, number of outputs = 16. $16 = 2^4 = 2^n$. Thus, number of inputs is 4.

9. A truth table with output columns numbered 0–15 may be for which type of decoder IC?

- a) Hexadecimal 1-of-16
- b) Dual octal outputs
- c) Binary-to-hexadecimal
- d) Hexadecimal-to-binary

Answer: a

Explanation: A binary decoder is a combinational logic circuit that decodes binary information from n -inputs to a maximum of 2^n outputs. A truth table with output columns numbered 0–15 may be for Hexadecimal 1-of-16. Because hexadecimal occupies less space in a system.

10. How can the active condition (HIGH or LOW) or the decoder output be determined from the logic symbol?

- a) A bubble indicates active-HIGH
- b) A bubble indicates active-LOW
- c) A triangle indicates active-HIGH
- d) A triangle indicates active-LOW

Answer: b

Explanation: A bubble indicates active-LOW in a decoder always. Enable pin of the decoder is usually active-LOW and is triggered on the input being at 0.

11. Which error detection method uses one's complement arithmetic?

- a) Simple parity check
- b) Two-dimensional parity check
- c) CRC
- d) Checksum

Answer: d

Explanation: A checksum is an error detection method used for the purpose of detecting errors that may have been incorporated during transmission. The checksum can be generated simply by adding bits. Hence, one's complement arithmetic uses checksum.

12. Which error detection method consists of just one redundant bit per data unit?

- a) Simple parity check
- b) Two-dimensional parity check
- c) CRC
- d) Checksum

Answer: a

Explanation: A parity checker is an error detection method used for the purpose of detecting errors that may have been incorporated during transmission. Simple parity check method consists of just one redundant bit per data unit. It is again classified as even parity and odd parity.

13. How many types of parity bits are found?

- a) 2
- b) 3
- c) 4

d) 1

Answer: a

Explanation: There are two types of parity bits, namely even parity and odd parity. In even parity, a 1 bit is added in order to make a group of data bits have even number of 1s. While, in odd parity, a 1 bit is added in order to make a group of data bits have odd number of 1s.

14. What is a parity bit?

- a) An error detection is achieved by adding an extra bit
- b) After addition, the carry is found
- c) Bit generated during data transmission
- d) After addition, the total number of bits

Answer: a

Explanation: A simple form of error detection is achieved by adding an extra bit to the transmitted word. The additional bit is known as parity bits.

15. The BCD number 101011 has _____ priority.

- a) Even
- b) Odd
- c) Both even and odd
- d) Undefined

Answer: a

Explanation: The given BCD number 101011 has even priority because it has an even number of 1's (i.e. 4).

16. Which error detection method involves polynomials?

- a) Simple parity check
- b) CRC
- c) Two-dimensional parity check
- d) Checksum

Answer: b

Explanation: Cyclic Redundancy Check(CRC) involves parity check polynomials. In the even parity case of CRC, the 1-bit is generated by checking the polynomial $x+1$.

17. The odd parity output of decimal number 9 is _____

- a) 0
- b) 1
- c) 1001
- d) 0011

Answer: b

Explanation: The odd parity output of decimal number 9 is 1 because the BCD number for 9 is 1001 and it has even number of 1's.

18. If odd parity is used for ASCII error detection, the number of 0s per 8-bit symbol is

- a) Indeterminate
- b) 42
- c) Even
- d) Odd

Answer: a

Explanation: Odd parity bit is 1 when the group of data bits consists of even number of 1s. So to make the group of data bits have odd number of 1s, 1 extra bit is added. If odd parity is used for ASCII error detection, the number of 0s per 8-bit symbol is indeterminate because it is applicable only for 6-bit symbol.

19. Which error detection method can detect a single-bit error?

- a) Simple parity check
- b) Two-dimensional parity check
- c) CRC
- d) Checksum

Answer: b

Explanation: A single-bit error can be detected by using two-dimensional parity check method. Since it converts the 4-bit number into 8-bit and count the number of one's.

20. Which gates are ideal for checking the parity bits?

- a) AND
- b) NAND
- c) EX-OR
- d) EX-NOR

Answer: c

Explanation: Exclusive-OR gates are ideal for checking the parity of a binary number because they produce an output when the input has an odd number of 1's. Therefore, an even-parity input to an EX-OR gate produces a low output, while an odd parity input produces a high output. While, in case of AND, it produces high output when all inputs are 1 else low. Whereas, NAND, does the opposite, by producing low output when all inputs are 1 else high.

7.) Code Converters

1. A code converter is a logic circuit that _____

- a) Inverts the given input
- b) Converts into decimal number
- c) Converts data of one type into another type
- d) Converts to octal

Answer: c

Explanation: A code converter is a logic circuit that changes data presented in one type of binary code to another type of binary code.

2. Use the weighting factors to convert the following BCD numbers to binary _____

0101 0011 & 0010 0110 1000

- a) 01010011 001001101000
- b) 11010100 100001100000
- c) 110101 100001100
- d) 101011 001100001

Answer: c

Explanation: Firstly, convert every 4 sets of binary to decimal from the given: 0101=5, 0011=3. Then convert 53 to binary, which will give 110101. Again, do the same with the next 4 set of binary digits.

3. The primary use for Gray code is _____

- a) Coded representation of a shaft's mechanical position
- b) Turning on/off software switches
- c) To represent the correct ASCII code to indicate the angular position of a shaft on rotating machinery
- d) To convert the angular position of a shaft on rotating machinery into hexadecimal code

Answer: a

Explanation: Gray code is useful because only one bit changes at a time, which is implemented easily in Coded representation of a shaft's mechanical position. In Gray Code, every sequence of successive bits differs by 1 bit only.

4. Code is a symbolic representation of _____

- a) Discrete information
- b) Continuous information
- c) Decimal information into binary
- d) Binary information into decimal

Answer: a

Explanation: Code is a symbolic representation of discrete information. Codes can be anything like numbers, letter or words, written in terms of group of symbols.

5. One way to convert BCD to binary using the hardware approach is _____

- a) With MSI IC circuits
- b) With a keyboard encoder
- c) With an ALU
- d) UART

Answer: a

Explanation: One way to convert BCD to binary using the hardware approach is MSI IC (i.e. medium scale integration) circuits.

6. Why is the Gray code more practical to use when coding the position of a rotating shaft?

- a) All digits change between counts
- b) Two digits change between counts
- c) Only one digit changes between counts
- d) Alternate digit changes between counts

Answer: c

Explanation: The Gray code is more practical to use when coding the position of a rotating shaft because only one digit changes between counts that is reflected to the next count.

7. Reflected binary code is also known as _____

- a) BCD code
- b) Binary code
- c) ASCII code
- d) Gray Code

Answer: d

Explanation: The reflected binary code is also known as gray code because one digit reflected to the next bit. In Gray Code, every sequence of successive bits differs by 1 bit only.

8. Why do we use gray codes?

- a) To count the no of bits changes
- b) To rotate a shaft
- c) Error correction
- d) Error Detection

Answer: c

Explanation: Today, Gray codes are widely used to facilitate error correction in digital communications such as digital terrestrial television and some cable TV systems.

9. Earlier, reflected binary codes were applied to _____

- a) Binary addition
- b) 2's complement
- c) Mathematical puzzles
- d) Binary multiplication

Answer: c

Explanation: The reflected binary code is also known as gray code because one digit reflected to the next bit. In Gray Code, every sequence of successive bits differs by 1 bit only. Reflected binary codes were applied to mathematical puzzles before they became known to engineers.

10. The binary representation of BCD number 00101001 (decimal 29) is _____

- a) 0011101
- b) 0110101
- c) 1101001
- d) 0101011

Answer: a

Explanation: The given BCD number 00101001 has three 1s. So, it can be rewritten as 0000001-1, 0001000-8, 0010100-20 and after addition, we get 0011101 as output.

11. Convert binary number into gray code: 100101.

- a) 101101
- b) 001110
- c) 110111
- d) 111001

Answer: c

Explanation: : Conversion from Binary To Gray Code:

1 (XOR) 0 (XOR) 0 (XOR) 1 (XOR) 0 (XOR) 1

↓ ↓ ↓ ↓ ↓

1 1 0 1 1 1

8.) Magnitude Comparator

1. All the comparisons made by comparator is done using _____

- a) 1 circuit
- b) 2 circuits
- c) 3 circuits
- d) 4 circuits

Answer: a

Explanation: A comparator is a combinational circuit that takes two numbers as input in binary form and results whether one input is greater, lesser or equal to the other input. Because, all the input is compared to each other, therefore it is possible only by using 1 circuit.

2. One that is not the outcome of magnitude comparator is _____

- a) $a > b$
- b) $a - b$
- c) $a < b$
- d) $a = b$

Answer: b

Explanation: A comparator is a combinational circuit that takes two numbers as input in binary form and results whether one input is greater, lesser or equal to the other input. In a digital comparator, only 3 outputs are possible (i.e. $A = B$, $A > B$, $A < B$). So, $a - b$ is an incorrect option.

3. If two numbers are not equal then binary variable will be _____

- a) 0
- b) 1
- c) A
- d) B

Answer: a

Explanation: A comparator is a combinational circuit that takes two numbers as input in binary form and results whether one input is greater, lesser or equal to the other input. In a digital comparator, only 3 outputs are possible (i.e. $A = B$, $A > B$, $A < B$). Other than this, the output will be 0.

4. How many inputs are required for a digital comparator?

- a) 1
- b) 2
- c) 3
- d) 4

Answer: b

Explanation: A comparator is a combinational circuit that takes two numbers as input in

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binary form and results whether one input is greater, lesser or equal to the other input. Thus, there are two inputs required for a digital comparator (i.e. A & B).

5. In a comparator, if we get input as $A > B$ then the output will be _____

- a) 1
- b) 0
- c) A
- d) B

Answer: a

Explanation: A comparator is a combinational circuit that takes two numbers as input in binary form and results whether one input is greater, lesser or equal to the other input. If $A > B$, it means that it satisfies one of the condition among three. Hence the output will be 1.

6. Which one is a basic comparator?

- a) XOR
- b) XNOR
- c) AND
- d) NAND

Answer: b

Explanation: Generally, an XNOR outputs high for even number of 1s or all 0s and outputs low for otherwise. Thus, an XNOR gate is a basic comparator, because its output is "1" only if its two input bits are equal.

7. Comparators are used in _____

- a) Memory
- b) CPU
- c) Motherboard
- d) Hard drive

Answer: b

Explanation: Comparators are used in the central processing unit (CPUs). Because all the arithmetic and logical operations are performed in the ALU(Arithmetic Logic Unit) part of the CPU.

8. A circuit that compares two numbers and determines their magnitude is called _____

- a) Height comparator
- b) Size comparator
- c) Comparator
- d) Magnitude comparator

Answer: d

Explanation: A comparator is a combinational circuit that takes two numbers as input in binary form and results whether one input is greater, lesser or equal to the other input. A circuit that compares two numbers and determine their magnitude is called magnitude comparator.

9. A procedure that specifies finite set of steps is called _____

- a) Algorithm
- b) Flow chart
- c) Chart
- d) Venn diagram

Answer: a

Explanation: A procedure that specifies finite set of steps is called algorithm, While a flowchart is a pictorial representation of the algorithm.

10. How many types of digital comparators are?

- a) 1
- b) 2
- c) 3
- d) 4

Answer: b

Explanation: There are two main types of Digital Comparator available and these are: Identity Comparator & Magnitude Comparator. Identity Comparator checks only the equality of the inputs and thus has one output terminal. While Magnitude Comparator checks for greater than, less than as well as equality of the inputs, and thus has 3 output terminals.

11. An identify comparator is defined as a digital comparator which has _____

- a) Only one output terminal
- b) Two output terminals
- c) Three output terminals
- d) No output terminal

Answer: a

Explanation: An Identity Comparator is a digital comparator that has only one output terminal for when $A = B$ either "HIGH" $A = B = 1$ or "LOW" $A = B = 0$.

12. A magnitude comparator is defined as a digital comparator which has _____

- a) Only one output terminal
- b) Two output terminals

- c) Three output terminals
- d) No output terminal

Answer: c

Explanation: A Magnitude Comparator is a digital comparator which has three output terminals, one each for equality, $A = B$ greater than, $A > B$ and less than $A < B$.

13. The purpose of a Digital Comparator is _____

- a) To convert analog input into digital
- b) To create different outputs
- c) To add a set of different numbers
- d) To compare a set of variables or unknown numbers

Answer: d

Explanation: A comparator is a combinational circuit that takes two numbers as input in binary form and results whether one input is greater, lesser or equal to the other input. The purpose of a Digital Comparator is to compare a set of variables or unknown numbers, for example A ($A_1, A_2, A_3, \dots A_n$, etc) against that of a constant or unknown value such as B ($B_1, B_2, B_3, \dots B_n$, etc) and produce an output condition or flag depending upon the result of the comparison.

14. TTL 74LS85 is a _____

- a) 1-bit digital comparator
- b) 4-bit magnitude comparator
- c) 8-bit magnitude comparator
- d) 8-bit word comparator

Answer: b

Explanation: TTL 74LS85 is a 4-bit magnitude comparator.
